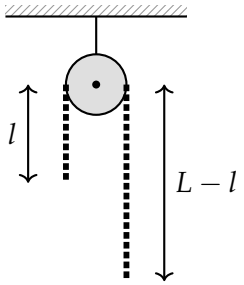
Integrated Programme – JEE MAIN Revision Sheet

NLM & Friction

Level 2 Sheet

Time to be taken: 60 Min Max

Q1. A uniform metal chain of mass m and length L passes over a massless and frictionless pulley. It is released from rest with a part of its length l is hanging on one side and rest of its length $L - l$ is hanging on the other side of the pulley. At a certain point of time, when $l = \frac{L}{x}$, the acceleration of the chain is $\frac{g}{2}$. The value of x is _____.



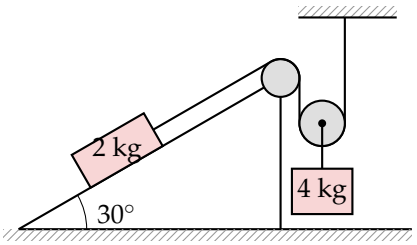
(A) 6 (B) 2 (C) 1.5 (D) 4

(Center of Mass Application)

Q2. A balloon has mass of 10 g in air. The air escapes from the balloon at a uniform rate with velocity 4.5 cm s^{-1}. If the balloon shrinks in 5 s completely. Then, the average force acting on that balloon will be (in dyne).

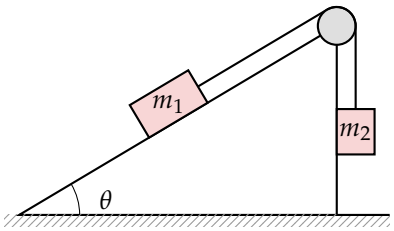
(A) 3 (B) 9 (C) 12 (D) 18

Q3. All surfaces shown in figure are assumed to be frictionless and the pulleys and the string are light. The acceleration of the block of mass 2 kg is:



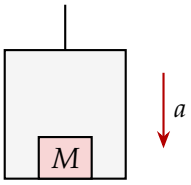
(Including Constraint and NLM)

Q4. Two bodies of masses $m_1 = 5$ kg and $m_2 = 3$ kg are connected by a light string going over a smooth light pulley on a smooth inclined plane as shown in the figure. The system is at rest. The force exerted by the inclined plane on the body of mass m_1 will be : [Take $g = 10$ m s^{-2}]



(A) 30 N (B) 40 N (C) 50 N (D) 60 N

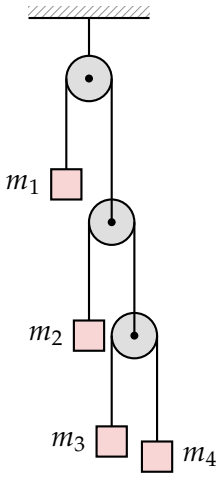
Q5. A block of mass M placed inside a box descends vertically with acceleration a. The block exerts a force equal to one-fourth of its weight on the floor of the box. The value of 'a' will be



(A) g (B) $\frac{3g}{4}$ (C) $\frac{g}{2}$ (D) $\frac{g}{4}$

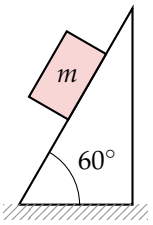
(Practice all lift questions)

Q6. In the arrangement shown in figure a_1, a_2, a_3 and a_4 are the accelerations of masses m_1, m_2, m_3 and m_4 respectively. Which of the following relation is true for this arrangement?



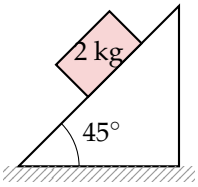
(A) $4a_1 + 2a_2 + a_3 + a_4 = 0$ (B) $a_1 + 4a_2 + 3a_3 + a_4 = 0$
(C) $a_1 + 4a_2 + 3a_3 + 2a_4 = 0$ (D) $2a_1 + 2a_2 + 3a_3 + a_4 = 0$

Q7. A cubic block of mass m is sliding down on an inclined plane at 60° with an acceleration of $\frac{g}{2}$, the value of coefficient of kinetic friction is



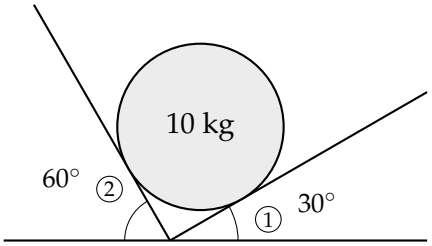
(A) $\sqrt{3} - 1$ (B) $\frac{\sqrt{3}}{2}$ (C) $\frac{\sqrt{2}}{3}$ (D) $1 - \frac{\sqrt{3}}{2}$

Q8. A 2 kg brick begins to slide over a surface which is inclined at an angle of 45° with respect to horizontal axis. The co-efficient of static friction between their surfaces is:

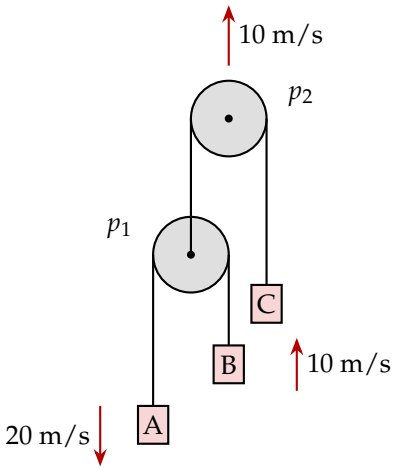


(A) 1.7 (B) $\frac{1}{\sqrt{3}}$ (C) 0.5 (D) 1

Q9. A solid sphere of mass 10 kg is placed over two smooth inclined planes as shown in figure. The normal reactions at 2 is $10x$ N. Find x ($g = 10\ \text{m/s}^2$)

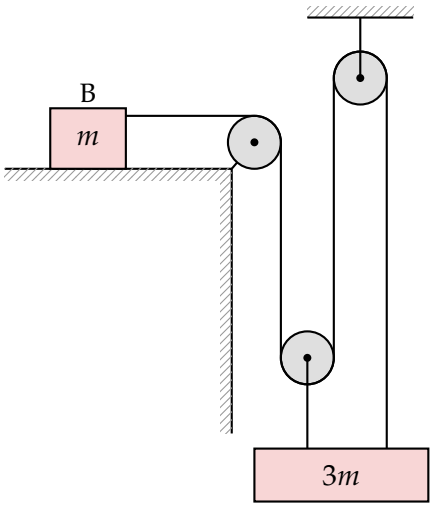


Q10. Velocities of blocks A, B and pulley p_2 are shown in figure. Find velocity of pulley p_1 and block C.

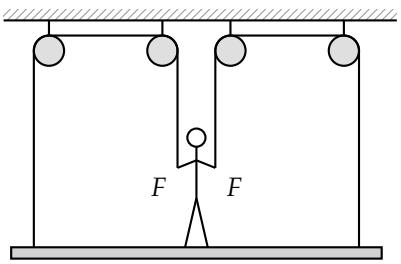


(revise Constraint Motion all types)

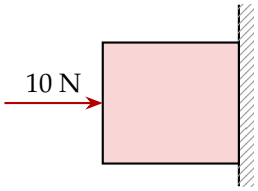
Q11. Find acceleration of both the blocks shown in the figure



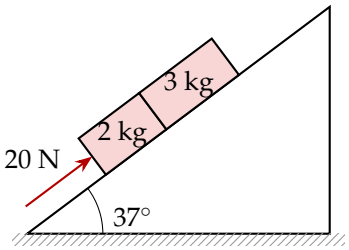
Q12. A painter of mass M stands on a platform of mass m and pulls himself up by two ropes which hang over pulley as shown. He pulls each rope with the force F and moves upward with uniform acceleration 'a'. Find 'a'



Q13. A horizontal force of 10 N is necessary to just hold a block stationary against a wall. The coefficient of friction between the block and the wall is 0.2. Find the weight (in Newton) of the block :



Q14. Find out the contact force between the 2 kg & 3 kg block as shown in figure



★ **KEEP GOING** ★

Hard problems are just easy problems in disguise. Stay calm, break the system into parts, and keep going — the questions that challenge you today are the ones that will make you unstoppable on exam day.